

JEE MAIN 2020 6th SEPT SHIFT 2**Physics**

Question 1. Find distance of centre of mass of solid hemisphere of radius 8 cm from centre

(1) 3 cm

(2) $\frac{3}{8}$ cm

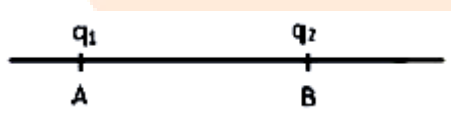
(3) 4 cm

(4) 2 cm

Ans. (1)

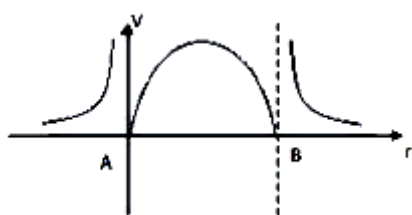
$$\text{Sol. } h_{\text{cm}} = \frac{3R}{8} = \frac{3 \times 8}{8} = 3\text{cm}$$

Question 2. Two charge $+q_1$ and $+q_2$ placed on x-axis at position A and B as shown in figure.

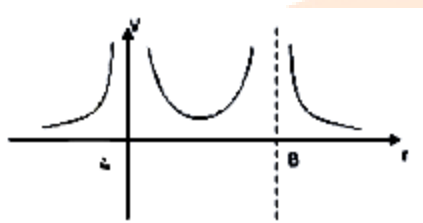


Which graph best represent the potential as a function of distance r from charge q_1

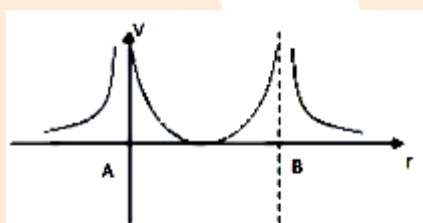
(1)



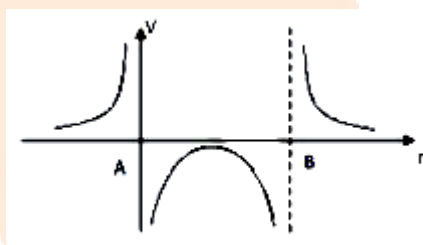
(2)



(3)



(4)



Ans. (2)

$$\text{Sol. } V = \frac{kq_1}{r} + \frac{kq_2}{x}$$

r = distance from charge q_1

x = distance from charge q_2

Question 3. Magnetic field of an electromagnetic wave is $\vec{B} = 12 \times 10^{-9} \sin(kx - \omega t) \hat{k} \text{ (T)}$. The equation of corresponding electric field should be:

- (1) $\vec{E} = 36 \sin(kx - \omega t) \hat{j} \text{ N/C}$
- (2) $\vec{E} = 3.6 \sin(kx - \omega t) \hat{j} \text{ N/C}$
- (3) $\vec{E} = 3.6 \sin(kx - \omega t) (-\hat{j}) \text{ N/C}$
- (4) $\vec{E} = 36 \sin(kx - \omega t) (-\hat{j}) \text{ N/C}$

Ans. (2)

Sol. $E_0 = B_0 \times C = 12 \times 10^{-9} \times 3 \times 10^8 = 3.6$

As the light is propagating in x direction

$$\& (\hat{E} \times \hat{B}) \parallel \hat{C}$$

$\therefore \vec{E}$ should be in $\hat{j}$ direction

So electric field $\vec{E} = E_0 \sin(kx - \omega t) \hat{j}$

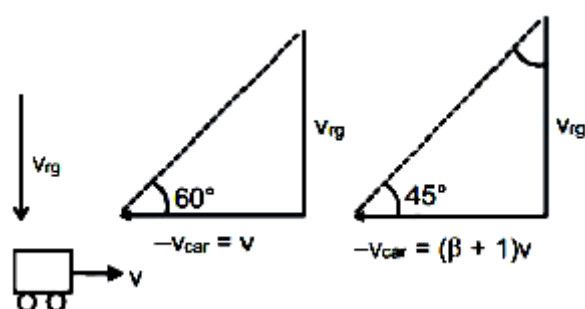
$$\vec{E} = 3.6 \sin(kx - \omega t) \hat{j}$$

Question 4. Rain is falling vertically when car is at rest. When car moves with speed v , rain appears at 60° with horizontal when car moves with speed $(\beta + 1)v$, rain appears at 45° with the horizontal. Find value of β :

- (1) 3
- (2) 0.732
- (3) 2
- (4) 0.5

Ans. (2)

Sol.



$$\tan 60^\circ = \frac{v_{rg}}{v} \dots (1)$$

$$\tan 45^\circ = \frac{v_{rg}}{(\beta + 1)v} \dots (2)$$

$$\sqrt{3}v = (\beta + 1)v$$

$$\beta = \sqrt{3} - 1 = 0.732$$

Question 5. An equi-concave lens has power P. Find power of plano-concave lens, when given lens is cut in such a way that two plano-concave lens are formed:

- (1) P
- (2) P/2

(3) $2P$

(4) $P/4$

Ans. (2)

Sol. $1/f = (n-1) [(-1/R) - (1/R)] = - (n-1)(2/R)$

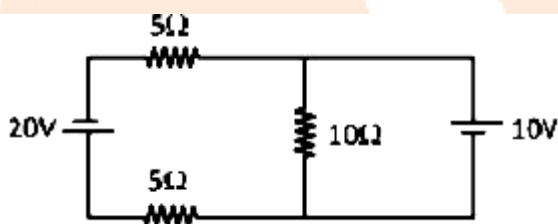
$$P = - R/2(n-1)$$

$$1/f' = (n-1) [(-1/R) - (0)] = -(n-1)/R$$

$$P' = -R/(n-1)$$

$$P' = P/2$$

Question 6. Find current in 10V cell.



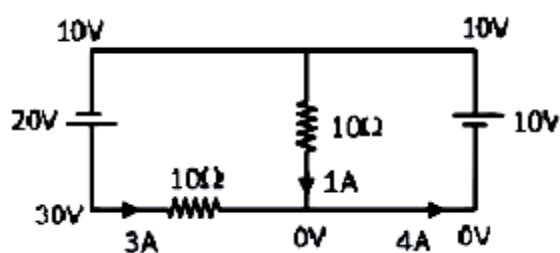
(1) 2A

(2) 3A

(3) 4A

(4) 8A

Ans. (3)



Sol.

So current in 10V cell is 4A

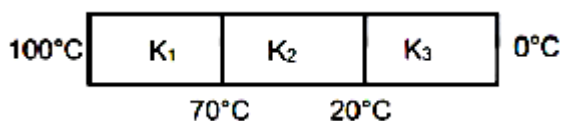
Question 7. In a electric circuit ammeter and voltmeter are used to verify ohm's law which of the following is true.

- (1) Ammeter connected in series and voltmeter connected in parallel
- (2) Voltmeter connected in series and ammeter connected in parallel
- (3) Both can be connected in parallel
- (4) Both can be connected in series

Ans. (1)

Sol. Theory Based

Question 8. Steady state temperature are shown in the diagram. Find ratio of thermal conductivity K_1/K_2 if length of all rods are same.



- (1) 3 : 2
- (2) 5 : 3
- (3) 4 : 2
- (4) 5 : 2

Ans. (2)

$$\text{Sol. } K_1 (100 - 70) = K_2 (70 - 20) = K_3 (20 - 0)$$

$$K_1 30 = K_2 (50) = K_3 (20)$$

$$\Rightarrow K_1 : K_2 : K_3 = 10 : 6 : 15$$

Question 9. A particle of mass m moving with speed v collides elastically with another particle of mass $2m$. Find speed of smaller mass after head on collision:

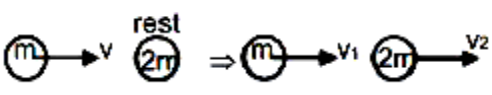
(1) $\frac{v}{3}$

(2) $-\frac{v}{3}$

(3) $\frac{2v}{3}$

(4) $-\frac{2v}{3}$

Ans. (2)

Sol. 

Apply conservation of linear momentum

$$mv = mv_1 + 2mv_2$$

$$\frac{v_2 - v_1}{v} = 1$$

Alternate:

$$v_1 = \frac{2m_2 u_2}{m_1 + m_2} + \frac{m_1 - m_2}{m_1 + m_2} u_1$$

$$= \frac{2 \times (2m)(0)}{3m} + \frac{(m - 2m)v}{3m} = -\frac{v}{3}$$

Question 10. Half-life of C19 is 5700 years. Find its decay constant.

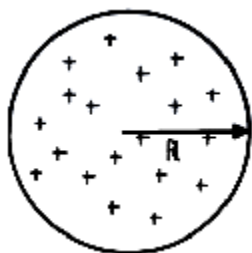
- (1) $12.16 \times 10^{-5} \text{ year}^{-1}$
- (2) $12.16 \times 10^{-3} \text{ year}^{-1}$
- (3) $8.52 \times 10^{-5} \text{ year}^{-1}$
- (4) $8.53 \times 10^{-3} \text{ year}^{-1}$

Ans. (1)

Sol. $\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.6932}{5700} = 12.16 \times 10^{-5} \text{ year}^{-1}$

Question 11. Charge density of a sphere of radius R is

$\rho = \frac{\rho_0}{r}$, where r is distance from centre of sphere. Total charge of sphere will be



- (1) $\rho_0 \pi R^2$

(2) $2\rho_0 \pi R^2$

(3) $3\rho_0 \pi R^2$

(4) $4\rho_0 \pi R^2$

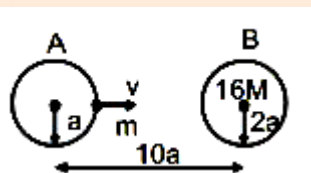
Ans. (2)

Sol. $Q = \int \rho dv = \int \rho 4\pi r^2 dr$

$$= \int_0^R \rho_0 4\pi r^2 dr = \int_0^R \rho_0 4\pi r dr$$

$$Q = 2\rho_0 \pi R^2$$

Question 12.



Two planets of mass M and $16M$ of radius a and $2a$ respectively, are at distance $10a$. Find minimum speed of a particle of mass m at surface of smaller planet so that it can reached from smaller planet to larger planet.

(1) $\sqrt{\frac{5}{9} \frac{GM}{a}}$

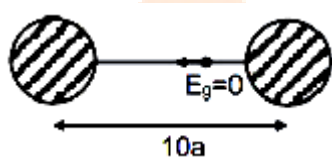
(2) $\sqrt{\frac{4}{9} \frac{GM}{a}}$

$$(3) \sqrt{\frac{3}{9} \frac{GM}{a}}$$

$$(4) \sqrt{\frac{2}{9} \frac{GM}{a}}$$

Ans. (1)

Sol.



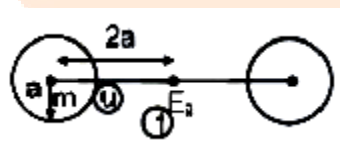
$$\frac{GM}{r^2} = \frac{16 GM}{(10a - r)^2}$$

$$\frac{1}{r} = \frac{4}{10a - r}$$

$$10a - r = 4r$$

$$5r = 10a$$

$$r = 2a$$



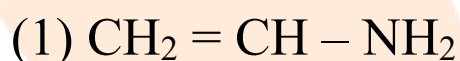
$$U_i + K_i = K_{\min} + U_f$$

$$-\frac{GMm}{a} - \frac{G16Mm}{9a} + \frac{1}{2}mv^2 = 0 - \frac{GMm}{2a} - \frac{G16Mm}{8a}$$

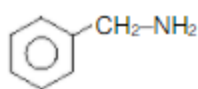
$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{GMm}{a} - \frac{GMm}{2a} + \frac{G16Mm}{9a} - \frac{G16Mm}{8a} \\ &= \frac{GMm}{2a} + \frac{16GMm}{a} \left(\frac{1}{9} - \frac{1}{8} \right) = \frac{GMm}{2a} - \frac{16GMm}{9 \times 8a} = \frac{GMm}{2a} - \frac{2GMm}{9a} \\ &= \frac{GMm}{a} \left[\frac{1}{2} - \frac{2}{9} \right]; \\ \frac{1}{2}mv^2 &= \frac{5}{18} \frac{GMm}{a} \\ v &= \sqrt{\frac{5}{9} \frac{GM}{a}}\end{aligned}$$

Chemistry

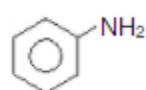
Question 13. Which of the following major product can be obtained from Gabriel phthalimide synthesis?



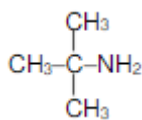
(2)



(3)



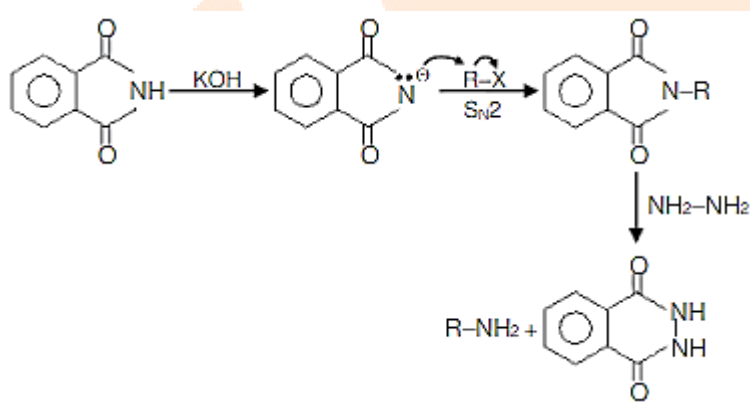
(4)



Answer: (2)

Solution:

From Gabriel phthalimide reaction, 1^o Amine can be prepared.



Question 14.

Column – 1 (Polymers)	Column – 2 (Monomers)
(I) Natural rubber	(a) Styrene
(II) Neoprene	(b) Isoprene
(III) Buna-N	(c) Chloroprene
(IV) Buna-S	(d) Acrylonitrile

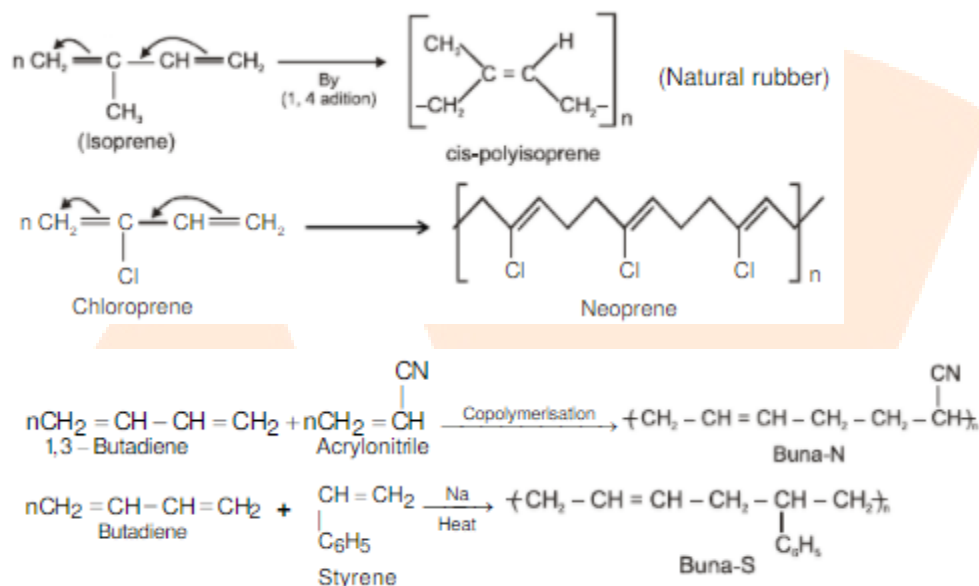
Which of the following combination is correct:

- (1) (I) – (b) ; (II) – (c) ; (III) – (d), (IV) – (a)
- (2) (I) – (c) ; (II) – (b) ; (III) – (d), (IV) – (a)
- (3) (I) – (b) ; (II) – (c) ; (III) – (a), (IV) – (d)

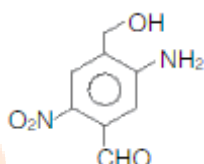
(4) (I) – (d) ; (II) – (b) ; (III) – (c), (IV) – (a)

Answer: (1)

Solution:



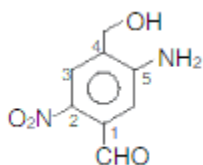
Question 15. The correct IUPAC name of given compound is:



- (1) 5-Formyl-2-(hydroxymethyl)-4-nitro aniline
- (2) 1-Amino-5-Formyl-2-(hydroxymethyl)-4-nitro benzene
- (3) 3-Amino-4-(hydroxymethyl)-6-nitro benzene carbaldehyde
- (4) 5-Amino-4-(hydroxymethyl)-2-nitro benzene carbaldehyde

Answer: (4)

Solution:



5-Amino-4-(hydroxymethyl)-2-nitro benzene carbaldehyde

Question 16.

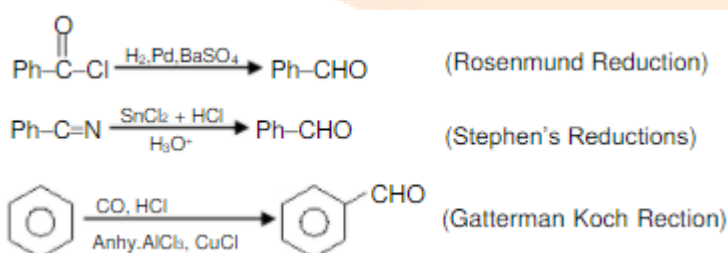
Column – 1	Column – 2
(I) Benzene	(a) H_2 -Pd-BaSO ₄ , Quinoline
(II) Benzonitrile	(b) CO, HCl/Anhydrous AlCl ₃ , CuCl
(III) Benzoyl chloride	(c) $\text{SnCl}_2 + \text{HCl}/\text{H}_3\text{O}^+$

Which combination gives benzaldehyde as a major product?

- (1) (I) – (a) ; (II) – (c) ; (III) – (b)
- (2) (I) – (b) ; (II) – (c) ; (III) – (a)
- (3) (I) – (c) ; (II) – (b) ; (III) – (a)
- (4) (I) – (c) ; (II) – (a) ; (III) – (b)

Answer: (2)

Solution:



Question 17.

Column – 1	Column – 2
(I) Lucas reagent	(a) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}/\text{aq. NaOH}$
(II) Dumas method	(b) $\text{Conc. H}_2\text{SO}_4/\Delta$
(III) Kjeldahl's method	(c) CuO/Δ
(IV) Heinsberg reagent	(d) An hy. $\text{ZnCl}_6/\text{Conc. HCl}$

Which of the following combination is correct:

- (1) (I) – (b) ; (II) – (c) ; (III) – (d), (IV) – (a)
- (2) (I) – (c) ; (II) – (b) ; (III) – (d), (IV) – (a)
- (3) (I) – (d) ; (II) – (c) ; (III) – (b), (IV) – (a)
- (4) (I) – (d) ; (II) – (b) ; (III) – (c), (IV) – (a)

Answer: (3)

Solution:

(I) Lucas reagent $\rightarrow$ Only $\text{ZnCl}_2/\text{Conc. HCl}$

(II) Dumas method $\rightarrow \text{CuO}/\Delta$

(III) Kjeldahl's method $\rightarrow \text{Conc. H}_2\text{SO}_4/\Delta$

(IV) Heinsberg reagent $\rightarrow \text{C}_6\text{H}_5\text{SO}_2\text{Cl} / \text{aq. NaOH}$

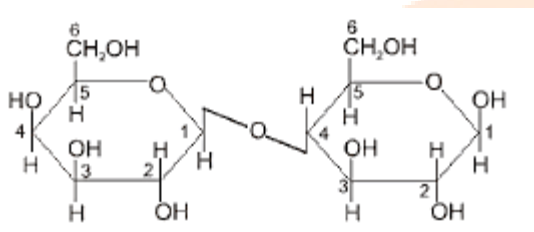
Question 18. Which statement is incorrect regarding Lactose?

- (1) It is composed of b-D(+)-Galactose & b-D(+)-Glucose
- (2) It is a reducing sugar
- (3) It is a disaccharides ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) having eight -OH groups

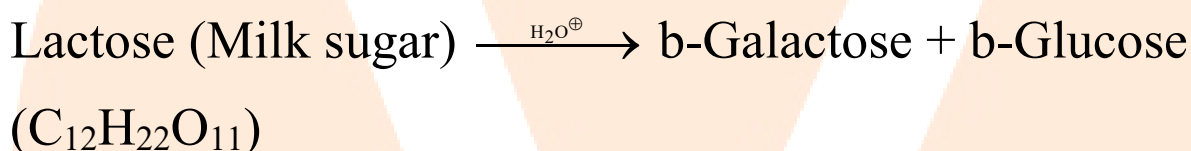
(4) It has glycosidic linkage of C1 of glucose and C₄ of galactose

Answer: (4)

Solution:

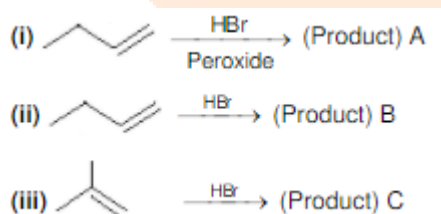


The linkage is between C – 1 of Galactose and C – 4 Glucose



It is hydrolysed by dilute acids or by the enzyme lactase, to an equimolecular mixture of D(+)-glucose and D(+)-galactose. Lactose is a reducing sugar.

Question 19. Complete the given reactions and compare the boiling points of their products?



(1) A > B > C

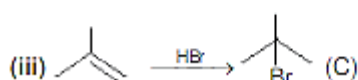
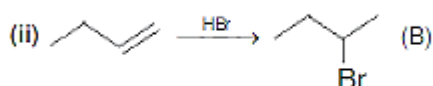
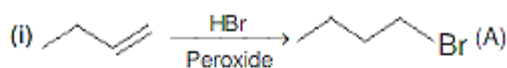
(2) A > C > B

(3) B > C > A

(4) $C > B > A$

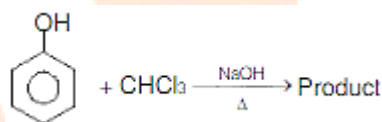
Answer: (1)

Solution:



The boiling points of isomeric halo alkanes decrease with increase in branching.

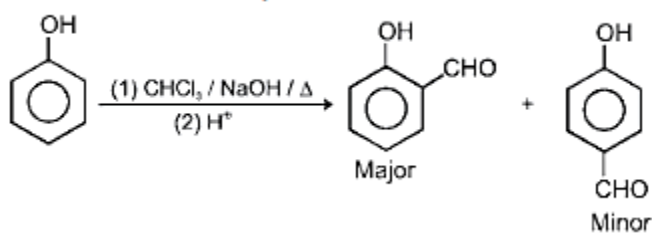
Question 20. Complete the given reaction and find the percentage weight of carbon in the product.



Answer: 68.85

Solution:

Reimer-Tiemann formylation reaction :



Molecular formula of product is $\text{C}_7\text{H}_6\text{O}_2$

$$\text{Percentage weight of carbon} = \left(\frac{84}{122} \times 100 \right) = 68.85\%$$

Question 21. For an equilibrium reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$; $K_c = 64$. What is the equilibrium constant the reaction $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{2}{3}\text{H}_2(\text{g})$

(1) $\frac{1}{8}$

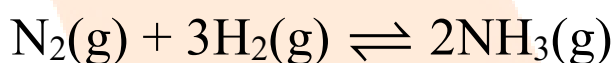
(2) $\frac{1}{4}$

(3) $\frac{1}{64}$

(4) $\frac{1}{2}$

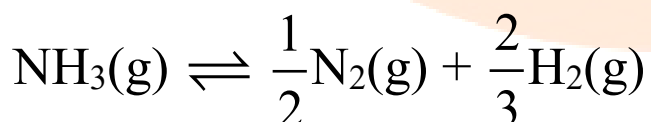
Answer: (1)

Solution:



$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = 64$$

For the reaction



$$K'_c = \frac{[\text{NH}_2]^{\frac{1}{2}} [\text{H}_2]^{\frac{3}{2}}}{[\text{NH}_3]} = \frac{1}{\sqrt{K_c}} = \frac{1}{\sqrt{64}} = \frac{1}{8}$$

Question 22. Which of the following is the correct electronic configuration in octahedral complex with d_4 configuration?

- (1) t_{2g}^4, e_g^0 ; $\Delta_0 < P$
- (2) t_{2g}^3, e_g^1 ; $\Delta_0 > P$
- (3) t_{2g}^2, e_g^2 ; $\Delta_0 < P$
- (4) t_{2g}^3, e_g^1 ; $\Delta_0 < P$

Answer: (4)

Solution:

For d_4 configuration if $\Delta_0 < P$ the electronic configuration is t_{2g}^3, e_g^1

Question 23. Which of the following method is used to produce di hydrogen of high degree purity ($> 99.95\%$) in maximum amount?

- (1) $\text{Zn} + \text{NaOH}$
- (2) with electrolysis of warm aqueous barium hydroxide solution between nickel electrodes
- (3) by electrolysis of brine solution
- (4) by electrolysis of alkaline water using platinum electrodes.

Answer: (2)

Solution:

Dihydrogen of high degree of purity ($> 99.95\%$) is obtained by the electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

Question 24. In a metal oxide, oxide ions crystallises in ccp lattice in which metal M_1 occupies 50% of octahedral voids and metal M_2 occupies 12.5% of tetrahedral voids. Then the oxidation states of metal M_1 and M_2 respectively are

(1) +2, +4

(2) +1, +3

(3) +3, +2

(4) +3, +1

Answer: (3)

Solution:

In the ccp lattice of oxide ions effective number of O^{2-} ions

$$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

In the ccp lattice,

No. of octahedral voids = 4

No. of tetrahedral voids = 8

$$\text{Given } M_1 \text{ metal atoms} = 4 \times \frac{50}{100} = 2$$

$$\text{Given } M_2 \text{ metal atoms} = 8 \times \frac{12.5}{100} = 1$$

\ Formula of the compound = $(M_1)_2(M_2)O_4$

\ Oxidation states of metals M_1 & M_2 respectively are +3 and +2.

Question 25. For the given concentration cell $Cu(s) | Cu^{2+}(C_2M) || Cu^{2+}(C_1M) | Cu(s)$

Gibb's energy (ΔG) is negative if:

(1) $C_2 = \frac{C_1}{\sqrt{2}}$

(2) $C_2 = 2C_1$

(3) $C_2 = \sqrt{2}C_1$

(4) $C_1 = C_2$

Answer: (2)

Solution:

For concentration cell $E_{\text{cell}} = 0$

Anode: $Cu(s) \rightarrow Cu^{2+}(aq)A$

Cathode: $\underline{Cu^{2+}(aq)C} \rightarrow \underline{Cu(s)}$

Overall: $Cu^{2+}(aq)C \rightarrow Cu^{2+}(aq)A$

As $\Delta G = -nF E_{\text{cell}}$

If $\Delta G = -ve$ then E_{cell} is positive

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.059}{2} \log \frac{C_2}{C_1}$$

$$E_{\text{cell}} = \frac{-0.059}{2} \log \frac{C_2}{C_1}$$

$$E_{\text{cell}} > 0 \Rightarrow C_2 < C_1$$

Question 26. Main constituent part of misch metal alloy is:

- (1) Lanthanids
- (2) Actinides
- (3) Transition metal
- (4) Both by Lanthanides and Actinides

Answer: (1)

Solution:

Misch metal consists of Lanthanide metal ($\approx 95\%$) and iron ($\approx 5\%$) and traces of S, C, Ca and Al.

Question 27. Distillation process is used for purification of

- (1) Zn
- (2) Ge
- (3) Ni
- (4) Co

Answer: (1)

Solution:

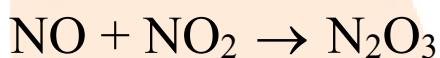
Zn, Cd & Hg are purified by fractional distillation process.

Question 28. What is product of following reaction, $\text{NO} + \text{NO}_2 \rightarrow$



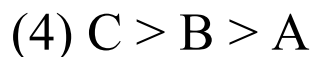
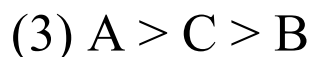
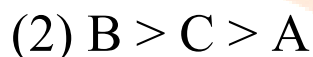
Answer: (2)

Solution:



Question 29. In 180 gram water 10 gram each of A, B and C are mixed separately, then correct order of vapour pressure of these solution is

(Given Molar masses of A, B and C are 100, 150 & 125 gram/mole respectively)



Answer: (2)

Solution:

Relative lowering in vapour pressure depends on no. of mole of solute greater the no. of mole of solute greater in RLVP and smaller will be vapour pressure.

So order of vapour pressure is $B > C > A$

Question 30. Average atomic mass of chlorine is 35.5 then the correct naturally occurring molar ratio of ^{35}Cl & ^{37}Cl is:

(1) 4 : 1

(2) 2 : 1

(3) 3 : 1

(4) 1 : 1

Answer: (3)

Solution:



Molar ratio x $1 - x$

$$M_{\text{avg.}} = 35 \times x + 37 (1 - x) = 35.5$$

$$35x + 37 - 37x = 35.5$$

$$2x = 1.5$$

$$x = \frac{3}{4}$$

So, ratio of $^{35}\text{Cl} : ^{37}\text{Cl} = 3 : 1$

Question 32. Match the compounds listed in Column – I with use in Column – II

	Column – I		Column – II
(i)	Ca(OH)_2	(a)	White wash
(ii)	$\text{CaSO}_4, \frac{1}{2} \text{H}_2\text{O}$	(b)	Antacid
(iii)	NaCl	(c)	Washing soda preparation
(iv)	Mg(OH)_2	(d)	In making moulds for plaster statues

(1) (i) – (b) ; (ii) – (d) ; (iii) – (c) ; (iv) – (a)

(2) (i) – (d) ; (ii) – (a) ; (iii) – (c) ; (iv) – (b)

(3) (i) – (a) ; (ii) – (d) ; (iii) – (c) ; (iv) – (b)

(4) (i) – (a) ; (ii) – (d) ; (iii) – (b) ; (iv) – (c)

Answer: (3)

Solution:

(1) Ca(OH)_2 is used in white wash.

(2) Plaster of paris used in making of molds for plaster statues

(3) NaCl is used in preparation of washing soda

(4) A suspension of Mg(OH)_2 in water is used in medicine as an antacid under name of milk of magnesia

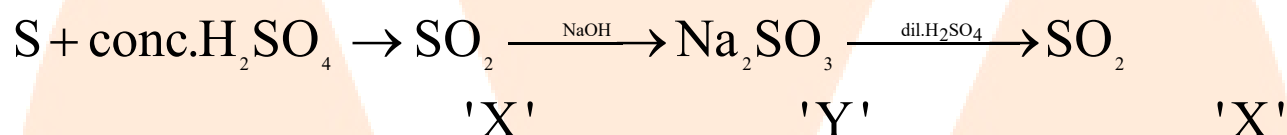
Question 33. Sulphur on reaction with conc. H_2SO_4 gives 'X' which on reaction with NaOH gives 'Y', Y on further

reaction with dil. H_2SO_4 give 'X' again. Then 'X' and 'Y' are respectively:

- (1) S, Na_2SO_3
- (2) SO_2 , Na_2SO_3
- (3) SO_2 , Na_2SO_4
- (4) SO_3 , Na_2SO_3

Answer: (2)

Solution:



Question 34. What is the atomic number of an element whose IUPAC name is Unnilunium?

Answer: 101

Solution:

According to IUPAC convention for naming of elements with atomic number more than 100, different digits are written in order and at the end sum is added. For digits following naming is used

0-nil

1-un

2-bi

3-tri

and so on...

Mathematic

Question 35. If α, β are the roots of equation $2x(2x + 1) = 1$ then $\beta = ?$

- (1) α^2
- (2) $-2\alpha(\alpha + 1)$
- (3) $2\alpha(\alpha + 1)$
- (4) $\alpha(\alpha - 1)$

Ans. (2)

Sol. Given equation is $2x(2x + 1) = 1 \Rightarrow 4x^2 + 2x - 1 = 0$

..(1)

roots of equation (1) are α and β

$$\therefore \alpha + \beta = -\frac{1}{2} \Rightarrow \beta = -\frac{1}{2} - \alpha \quad \dots(2)$$

$$\text{and } 4\alpha^2 + 2\alpha - 1 = 0 \Rightarrow \alpha^2 = \frac{1}{4} - \frac{\alpha}{2} \quad \dots(3)$$

$$\text{Now, } -2\alpha(\alpha + 1) = -2\alpha^2 - 2\alpha$$

$$= -2\left(\frac{1}{4} - \frac{\alpha}{2}\right) - 2\alpha = -\frac{1}{2} - \alpha = \beta.$$

Question 36. A plane intersects the x, y, z axis at A, B, C respectively. If G(1,1,2) is centroid of ΔABC , then the equation of the line perpendicular to plane and passing through G is

(1) $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-2}{2}$

(2) $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-2}{2}$

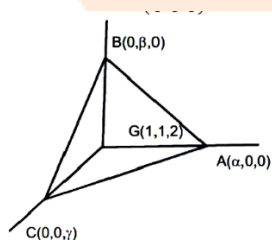
(3) $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$

(4) $\frac{x-1}{3} = \frac{y-1}{3} = \frac{z-2}{6}$

Ans. (3)

Sol. Let $A(\alpha, 0, 0)$, $B(0, \beta, 0)$, $C(0, 0, \lambda)$ then

$$G\left(\frac{\alpha}{3}, \frac{\beta}{3}, \frac{\lambda}{3}\right) \equiv (1, 1, 2)$$



$$\alpha = 3, \beta = 3, \lambda = 6$$

$$\therefore \text{equation of plane is } \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\lambda} = 1$$

$$\Rightarrow \frac{x}{3} + \frac{y}{3} + \frac{z}{6} = 1$$

$$\Rightarrow 2x + 2y + z = 6$$

$$\therefore \text{required line } \frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$$

Question 37. Total number of words (with or without meaning) from letters of word 'LETTER' if no two vowels are together

(1) 100

(2) 110

(3) 120

(4) 180

Ans. (3)

Sol. Consonants are L, T, T, R

Vowels are E, E,

Total number of word (with or without meaning) from

letters of word 'LETTER' = $\frac{6!}{2!2!} = 180$

Total number of words (with or without meaning) from

letters of word 'LETTER' if vowels are together = $\frac{5!}{2!} = 60$

$\therefore$ Required = $180 - 60 = 120$

Question 38. If the constant terms in the expansion of

$$\left(\sqrt{x} - \frac{K}{x^2}\right)^{10} \text{ is } 405 \text{ then } |K| = ?$$

(1) 9

(2) $\frac{9}{2}$

(3) 3

(4) 6

Ans. (3)

$$\text{Sol. } T_{r+1} = {}^{10}C_r \cdot \left(\frac{-K}{x^2}\right)^r (\sqrt{x})^{10-r}$$

$$= {}^{10}C_r \cdot (-K)^r \cdot x^{5-\frac{5r}{2}}$$

$$\text{For constant term} \Rightarrow 5 - \frac{5r}{2} = 0 \Rightarrow r = 2$$

$$\Rightarrow T_3 = {}^{10}C_2 \cdot K^2 = 405$$

$$\Rightarrow \frac{10(9)}{2} K^2 = 405$$

$$\Rightarrow K^2 = 9 \Rightarrow |K| = 3$$

Question 39. Centre of a circle passing through point (0, 1) and touching the curve $y = x^2$ at (2, 4) is

$$(1) \left(\frac{16}{5}, \frac{53}{10} \right)$$

$$(2) \left(\frac{-16}{5}, \frac{53}{10} \right)$$

$$(3) \left(\frac{-16}{5}, \frac{-53}{10} \right)$$

$$(4) \left(\frac{16}{5}, \frac{-53}{10} \right)$$

Ans. (2)

Sol. $y = x^2$, (2, 4)

Tangent at (2, 4) is

$$\frac{1}{2}(y + 4) = 2x$$

$$y + 4 = 4x \Rightarrow 4x - y - 4 = 0$$

$$\text{Equation of circle } (x - 2)^2 + (y - 4)^2 + \lambda(4x - y - 4) = 0$$

It passes through (0, 1)

$$\therefore 4 + 9 + \lambda(0 - 1 - 4) = 0$$

$$13 = 5\lambda \Rightarrow \lambda = \frac{13}{5}$$

$$\therefore \text{circle is } x^2 - 4x + 4 + y^2 - 8y + 16 + \frac{13}{5}(4x - y - 4) = 0$$

$$\Rightarrow x^2 + y^2 + \left(\frac{52}{5} - 4\right)x - \left(8 + \frac{13}{5}\right)y + 20 - \frac{52}{5} = 0$$

$$\Rightarrow x^2 + y^2 + \frac{32}{5}x - \frac{53}{5}y + \frac{48}{5} = 0$$

$$\therefore \text{centre is } \left(-\frac{16}{5}, \frac{53}{10}\right)$$

Question 40. If $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ and $\beta = A^4 + A$ then determinant of $\beta = ?$

(1) $2 - 2.\cos 3\theta$

(2) $2 + 3.\cos 2\theta$

(3) $3 + \cos 2\theta$

(4) $2 + 2.\cos 3\theta$

Ans. (4)

$$\text{Sol. } A^2 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$A^4 = \begin{bmatrix} \cos 4\theta & \sin 4\theta \\ -\sin 4\theta & \cos 4\theta \end{bmatrix}$$

$$\beta = \begin{bmatrix} \cos 4\theta & \sin 4\theta \\ -\sin 4\theta & \cos 4\theta \end{bmatrix} + \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 4\theta + \cos \theta & \sin 4\theta + \sin \theta \\ -(\sin 4\theta + \sin \theta) & \cos 4\theta + \cos \theta \end{bmatrix}$$

$$|\beta| = (\cos 4\theta + \cos \theta)^2 + (\sin 4\theta + \sin \theta)^2$$

$$= 2 + 2(\cos 4\theta \cdot \cos \theta + \sin 4\theta \cdot \sin \theta)$$

$$= 2 + 2\cos(4\theta - \theta)$$

$$= 2 + 2\cos 3\theta.$$

Question 41. Contrapositive of “If $n^3 - 1$ is even then n is odd” is

- (1) If n is odd then $n^3 - 1$ is not even
- (2) If n is not odd then $n^3 - 1$ is even
- (3) If n is not odd then $n^3 - 1$ is not even
- (4) If n is odd then $n^3 - 1$ is even

Ans. (3)

Sol. $p = n^3 - 1$ is even, $q : n$ is odd

Contrapositive of $p \rightarrow q = \sim q \rightarrow \sim p$

$\Rightarrow$ “If n is not odd then $n^3 - 1$ is not even”

Question 42. If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two arithmetic progression with common difference of 2nd

is two more than that of first and $b_{100} = a_{70}$, $a_{100} = -399$, $a_{40} = -159$ then the value of b_1 is

(1) -51

(2) -61

(3) -81

(4) 81

Ans. (3)

Sol. Let $a_1, a_1 + d, a_1 + 2d \dots$ first A. P.

$$a_{40} = a_1 + 39d = -159 \dots (1)$$

$$a_{100} = a_1 + 99d = -399 \dots (2)$$

from equation (1) and (2)

$$d = -4, a_1 = -3$$

$$\text{Now, } b_{100} = a_{70}$$

$$\Rightarrow b_1 + 99D = a_1 + 69d$$

$$b_1 + 99 \times (-2) = -3 + 69 \times (-4) \text{ (According to question } D = d + 2)$$

$$\Rightarrow b_1 = -81.$$

Question 43. If the angle of elevation of the top of a summit is 45° and a person climbs at an inclination of 30° upto 1km, where the angle of elevation of top becomes 60° , then height of the summit is

$$(1) \frac{1}{\sqrt{3}+1} \text{ km}$$

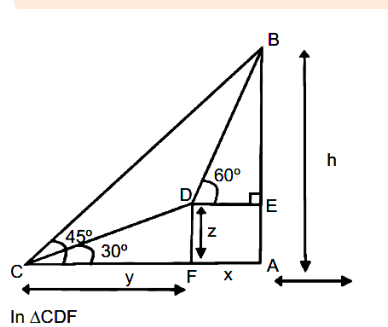
$$(2) \frac{\sqrt{3}-2}{2} \text{ km}$$

$$(3) \frac{\sqrt{3}+1}{2} \text{ km}$$

$$(4) \frac{3}{\sqrt{3}-1} \text{ km}$$

Ans. (3)

Sol.



In $\triangle CDF$

$$\sin 30^\circ = \frac{z}{1} \left[CD = 1 \text{ km (given)} \right]$$

$$z = \frac{1}{2} \quad \dots(1)$$

$$\cos 30^\circ = \frac{y}{1} \Rightarrow \frac{\sqrt{3}}{2} \quad \dots(1)$$

Now in $\triangle ABC$

$$\tan 45^\circ = \frac{h}{x+y}$$

$$\Rightarrow h = x + y$$

$$\Rightarrow x = h - \frac{\sqrt{3}}{2} \quad \dots(2)$$

Now in $\triangle BDE$

$$\tan 60^\circ = \frac{h-z}{x}$$

$$\sqrt{3}x = h - \frac{1}{2}$$

$$\Rightarrow \sqrt{3} \left(h - \frac{\sqrt{3}}{2} \right) = h - \frac{1}{2}$$

$$h = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2} \text{ km}$$

Question 44. If ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are frequencies of $n+1$ observations $1, 2, 22, \dots, 2_n$ such that mean is $\frac{729}{2^n}$ then value of n is:

Ans. (06.00)

Sol.

xi (observation)	1	2	22	...	2n
fi (frequency)	nC_0	nC_1	nC_2	...	nC_n

$$\begin{aligned}
 \bar{x} &= \frac{\sum f_i x_i}{\sum f_i} \\
 &= \frac{1 \times {}^nC_0 + 2 \times {}^nC_1 + 2^2 \times {}^nC_2 + \dots + 2^n \times {}^nC_n}{{}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n} \\
 &= \frac{(1+2)^n}{(1+1)^n} = \frac{3^n}{2^n} = \frac{729}{2^n} \\
 \Rightarrow 3^n &= 3^6 \\
 \Rightarrow n &= 6.
 \end{aligned}$$

Question 45. Area bounded by curves $y = x^2 - 1$ and $y = 1 - x^2$ is

(1) $\frac{1}{3}$

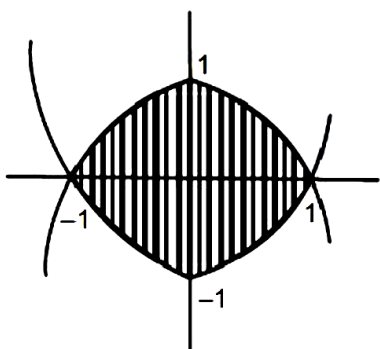
(2) $\frac{8}{3}$

(3) $\frac{4}{3}$

$$(4) \frac{2}{3}$$

Ans. (2)

Sol.



Given curves are $y = x^2 - 1$ and $y = 1 - x^2$

So, intersection points are $(\pm 1, 0)$ bounded area =

$$4 \int_0^1 (1 - x^2) dx = 4 \left[x - \frac{x^3}{3} \right]_0^1 = 4 \left(1 - \frac{1}{3} \right) = \frac{8}{3} \text{ sq. units}$$